

# Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: medium

**Benchmark:** HOPE\_DAC | **Context:** Global NLP/AI Research Community — Computational Mental Health (HOPE Benchmark Assessment)

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

HOPE's task taxonomy targets dialogue-act classification in dyadic counseling, explicitly distinguished from goal-oriented and chit-chat dialogue [Q7, Q8, Q9], with the well-defined objective of assigning one DAC label per utterance [Q22, Q58]. For the elicited use case — individual counseling-style chatbot conversations — this coverage is appropriate, and the elicitation explicitly affirms that 'individual counseling-style conversations' are well covered. However, the taxonomy is restricted to individual dyadic sessions [Q26], with no sub-task disaggregation for group therapy, crisis intervention, or culturally adapted modalities (e.g., spiritually grounded counseling, collectivist distress framing). The elicitation marks IO as MODERATE priority and treats cultural adaptation as deferred future work, so this is a known but accepted gap rather than a fundamental misalignment. The baseline architecture comparison and ablations [Q79, Q80, Q115] cover utterance, local context, global context, and speaker-awareness signals, which is structurally appropriate for the construct.

| ID   | Evidence                                                                                                                                                                                                                                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q7   | "Unlike general goal-oriented dialogues, a conversation between a patient and a therapist is considerably implicit, though the objective of the conversation is quite apparent." (p.1)                                                         |
| Q8   | "The nature of conversations in a social counselling setting is particularly distinct as compared to a conventional chit-chat or goal-oriented conversations." (p.1)                                                                           |
| Q9   | "It follows a pattern which is different from both goal-oriented and general chit-chat based conversations." (p.1)                                                                                                                             |
| Q22  | "In this section, we present our dialogue-act classification dataset, called HOPE. In total, we annotated ~ 12.9K utterances with 12 dialogue-act labels carefully designed to cater to the requirements of a counseling session." (p.3)       |
| Q26  | "The data collection process provides us 12.9K utterances from 212 counseling therapy sessions – all of them are dyadic conversations only." (p.3)                                                                                             |
| Q58  | "The objective of SPARTA is to assign a correct dialogue-act label $y_t$ to every utterance $U_t$ in the dialogue." (p.5)                                                                                                                      |
| Q79  | "Based on the type of modelling, we categorize the baselines into three groups – utterance-driven ( $U_t$ ), utterance+global context driven ( $U_t + GC$ ), and utterance + global context + speaker-aware driven ( $U_t + GC + SA$ )." (p.7) |
| Q80  | "Comparatively, SPARTA incorporates local context in addition to utterance, global context, and speaker dynamics ( $U_t + LC + GC + SA$ )." (p.7)                                                                                              |
| Q115 | "The first set of results (i.e., SPARTA-BS or the baseline version of SPARTA) in Table 7 represents the absence of local context in SPARTA." (p.11)                                                                                            |

## Strengths

- Task framing explicitly recognizes counseling dialogue as distinct from goal-oriented and chit-chat conversations [Q7, Q8, Q9], avoiding category misalignment with general-purpose DA schemes.
- Ablation design [Q115, Q118, Q125] systematically tests local context, global context, and speaker awareness — structurally relevant signals for counseling DAC.
- Coverage of individual dyadic CBT-compatible modalities aligns with the elicited primary use case (Q1/A1).

| ID | Evidence                                                                                                                                |
|----|-----------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | "We develop a novel dataset, named HOPE, to provide a platform for the dialogue-act classification in counselling conversations." (p.1) |

|             |                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q7</b>   | “Unlike general goal-oriented dialogues, a conversation between a patient and a therapist is considerably implicit, though the objective of the conversation is quite apparent.” (p.1)                                 |
| <b>Q8</b>   | “The nature of conversations in a social counselling setting is particularly distinct as compared to a conventional chit-chat or goal-oriented conversations.” (p.1)                                                   |
| <b>Q9</b>   | “It follows a pattern which is different from both goal-oriented and general chit-chat based conversations.” (p.1)                                                                                                     |
| <b>Q115</b> | “The first set of results (i.e., SPARTA-BS or the baseline version of SPARTA) in Table 7 represents the absence of local context in SPARTA.” (p.11)                                                                    |
| <b>Q118</b> | “Dialogue-act labels in a counseling based conversation not only depend on the abstract information of the dialogue but also on the utterances in the immediate vicinity of the current utterance.” (p.11)             |
| <b>Q125</b> | “This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances.” (p.11) |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                            |
|-------------|----------------------------------------------------------------|
| <b>IO-1</b> | Identify test case categories required for regional deployment |
| <b>IO-2</b> | Check if taxonomy omits regionally relevant categories         |
| <b>IO-3</b> | Check if taxonomy includes irrelevant categories               |
| <b>IO-4</b> | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** For the target context, required categories are dialogue-act types in individual counseling dyads — covered by HOPE's framing [Q22, Q58]. Culturally adapted modality categories (e.g., spiritually grounded counseling, community healing frameworks) would also be relevant for downstream chatbot deployment per the regional context, but are explicitly out of scope.

**IO-2:** Yes — the taxonomy omits sub-categories for crisis intervention, group therapy, motivational interviewing as a distinct modality, and culturally adapted therapeutic frameworks. The elicitation (A1) treats this as deferred future work rather than a current failure. No published culturally adapted DA taxonomy for counseling NLP exists [WEB-23], confirming this is a field-wide gap.

**IO-3:** No clearly irrelevant categories identified for the target context. The 12 DA categories [Q33–Q44] are general communicative-intent labels that plausibly apply across counseling sub-types.

**IO-4:** Documented gap: absence of culturally adapted counseling modality categories [WEB-23] and absence of crisis/group sub-task disaggregation. INSUFFICIENT DOCUMENTATION on per-modality session counts within HOPE — would need session-level metadata tagging therapy sub-type.

| ID         | Evidence                                                                                                                                                                                                                                 |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q22</b> | “In this section, we present our dialogue-act classification dataset, called HOPE. In total, we annotated ~ 12.9K utterances with 12 dialogue-act labels carefully designed to cater to the requirements of a counseling session.” (p.3) |
| <b>Q26</b> | “The data collection process provides us 12.9K utterances from 212 counseling therapy sessions – all of them are dyadic conversations only.” (p.3)                                                                                       |
| <b>Q33</b> | “Information Request (IRQ): This label is used as a request for some information, e.g., ‘Tell me your name.’” (p.4)                                                                                                                      |
| <b>Q34</b> | “Yes/No Question (YNQ): The YNQ label is similar to IRQ; however, the expected response is a trivial yes or no answer.” (p.4)                                                                                                            |
| <b>Q35</b> | “Clarification Request (CRQ): This label is assigned to those utterances in which a speaker usually asks the therapist for further clarification about topic that is being currently discussed.” (p.4)                                   |
| <b>Q36</b> | “Opinion Request (ORQ): The ORQ label is used when the speaker seeks opinions of the listener.” (p.4)                                                                                                                                    |
| <b>Q37</b> | “Clarification Delivery (CD): This label is used when the speaker provides further clarifications about a topic/entity under discussion.” (p.4)                                                                                          |
| <b>Q38</b> | “Information Delivery (ID): When the speaker provides some factual objective information about herself.” (p.4)                                                                                                                           |

|               |                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q39</b>    | “Positive Answer (PA): These labels are used when the utterance is an answer in the form of a simple yes to a question that was previously uttered.” (p.4)                                                                         |
| <b>Q40</b>    | “Negative Answer(NA): It is used when the utterance is an answer of the form of a simple No to a question asked earlier.” (p.4)                                                                                                    |
| <b>Q41</b>    | “Opinion Delivery (OD): When the speaker explicitly lists out her/his opinions.” (p.4)                                                                                                                                             |
| <b>Q42</b>    | “Greeting (GT): Each session usually starts with greeting by one speaker and an appropriate response from the other.” (p.4)                                                                                                        |
| <b>Q43</b>    | “Acknowledgment (ACK): In normal conversation, very often, we utter (e.g., 'Yeah! You are right.') to acknowledge the other speaker or to show our agreement without an explicit information request, question, or command.” (p.4) |
| <b>Q44</b>    | “General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation.” (p.4)                       |
| <b>Q58</b>    | “The objective of SPARTA is to assign a correct dialogue-act label $y_t$ to every utterance $U_t$ in the dialogue.” (p.5)                                                                                                          |
| <b>WEB-15</b> | AnnoMI uses 10 annotators with post-annotation survey, demonstrating richer annotator documentation is achievable ( <a href="https://www.mdpi.com/1999-5903/15/3/110">https://www.mdpi.com/1999-5903/15/3/110</a> )                |
| <b>WEB-23</b> | JMIR Mental Health (2025) confirms 'limited attention to linguistic or cultural adaptation' in LLM mental health studies ( <a href="https://mental.jmir.org/2025/1/e78410">https://mental.jmir.org/2025/1/e78410</a> )             |

## Information Gaps

- Per-modality session counts (CBT vs. child vs. family vs. substance use) within HOPE are not documented in retrievable sources.
- No empirical evidence of how the taxonomy performs across counseling sub-types.

## Requires Expert Verification

- Whether the 12-label scheme adequately captures dialogue acts in crisis intervention or culturally adapted therapeutic sessions — would require clinical and cross-cultural psychology expert review.

## Remediation

**Gap 1: Per-modality session counts (CBT, child, family, substance use) are not documented within HOPE, preventing modality-specific validity assessment.**

**Recommendation:** Publish session-level metadata tagging therapy sub-type and report performance disaggregated by sub-modality. Where coverage is thin (e.g., substance use), explicitly caveat downstream chatbot use in those modalities.